

UFO Sightings

Sistemas de Informação 1

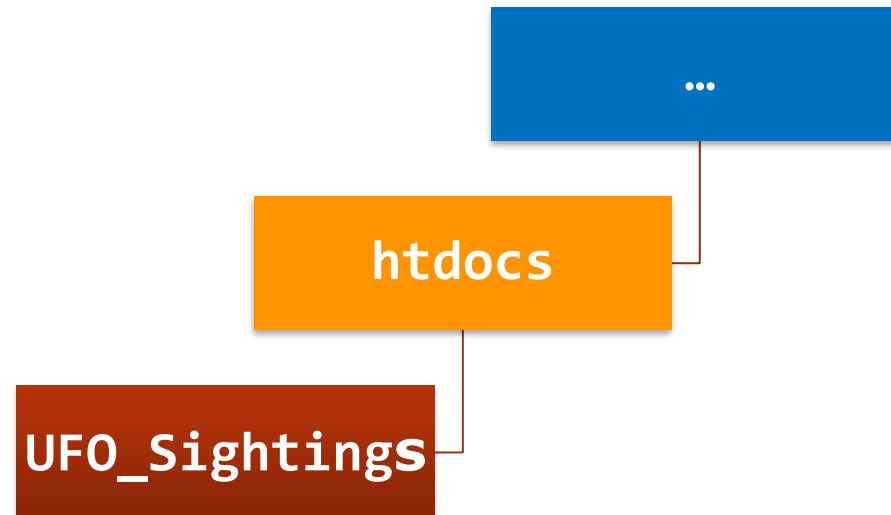
Setup Environment

- On Windows, you can download and install XAMPP.
 - XAMPP is a very easy to install Apache Distribution for Linux, Solaris, Windows, and Mac OS X.
 - The package includes the Apache web server, MySQL, PHP, Perl, a FTP server and phpMyAdmin.
 - Url: <https://sourceforge.net/projects/xampp/>
- On MacOS, you can download and install MAMP
 - MAMP is a free local server environment with Apache, Nginx, PHP, and MySQL.
 - Url: <https://www.mamp.info/en/mac/>



Create the project

- Go to **htdocs** folder
 - This is where the web server expects to find the web site
 - On Windows: “**.../xampp/htdocs**”
 - On MacOs: “**/Application/MAMP/htdocs**”
- Create a folder named **UFO_Sightings**

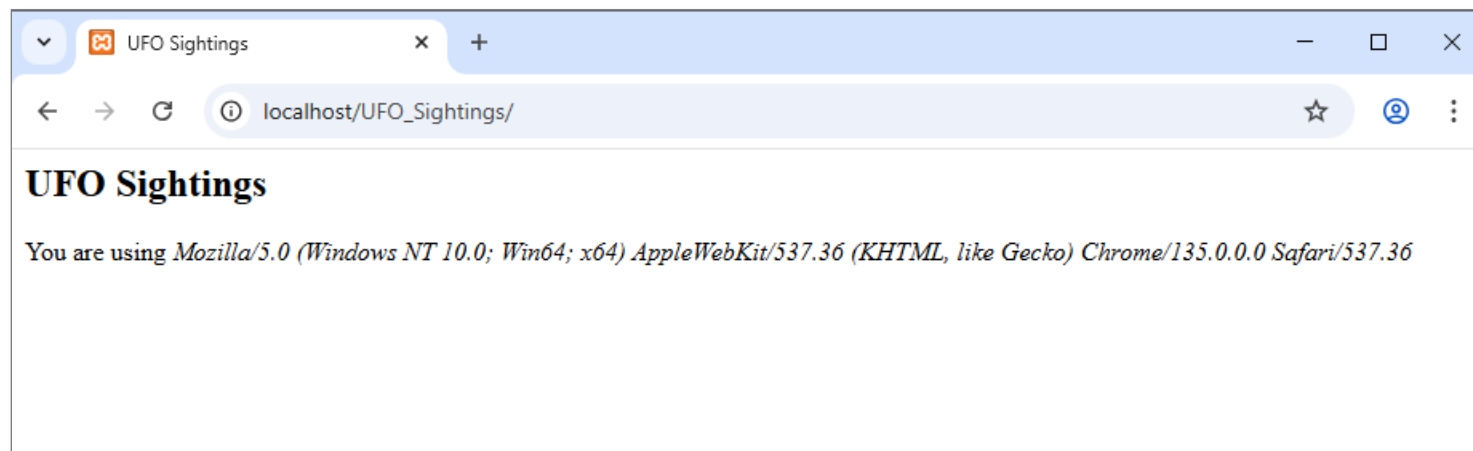
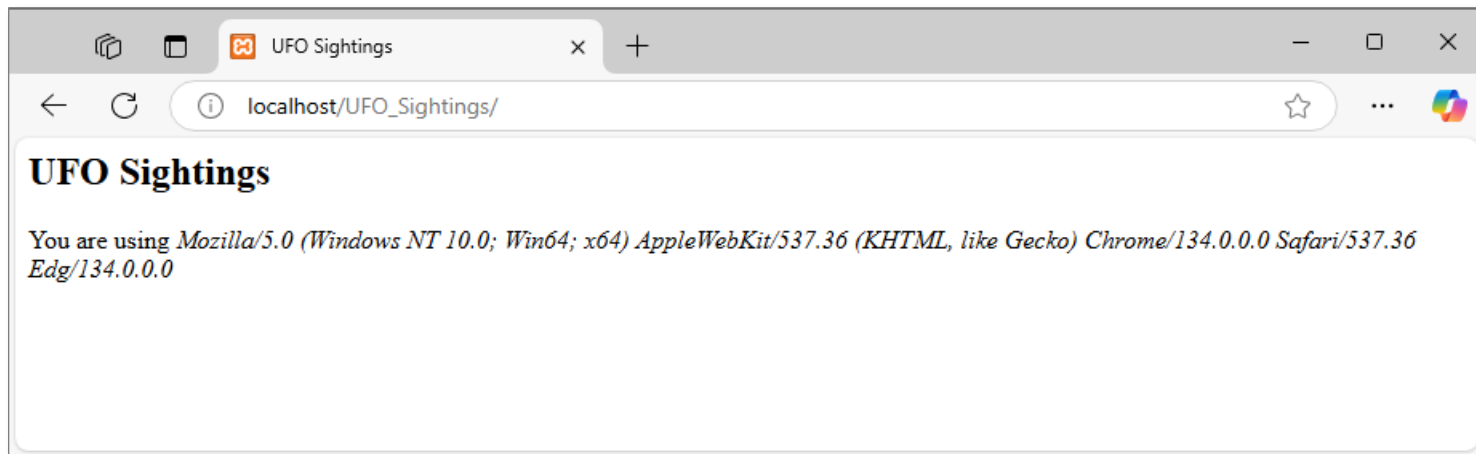


Create index.php file

- Create the **UFO_Sightings/index.php** file

```
<html>
  <head><title>UFO Sightings</title></head>
  <body>
    <h2>UFO Sightings</h2>
    <?php
      $browser = $_SERVER['HTTP_USER_AGENT'];
      echo "<p>You are using <i>$browser</i></p>";
    ?>
  </body>
</html>
```

- Start web server
- Open **http://localhost:????/UFO_Sightings**

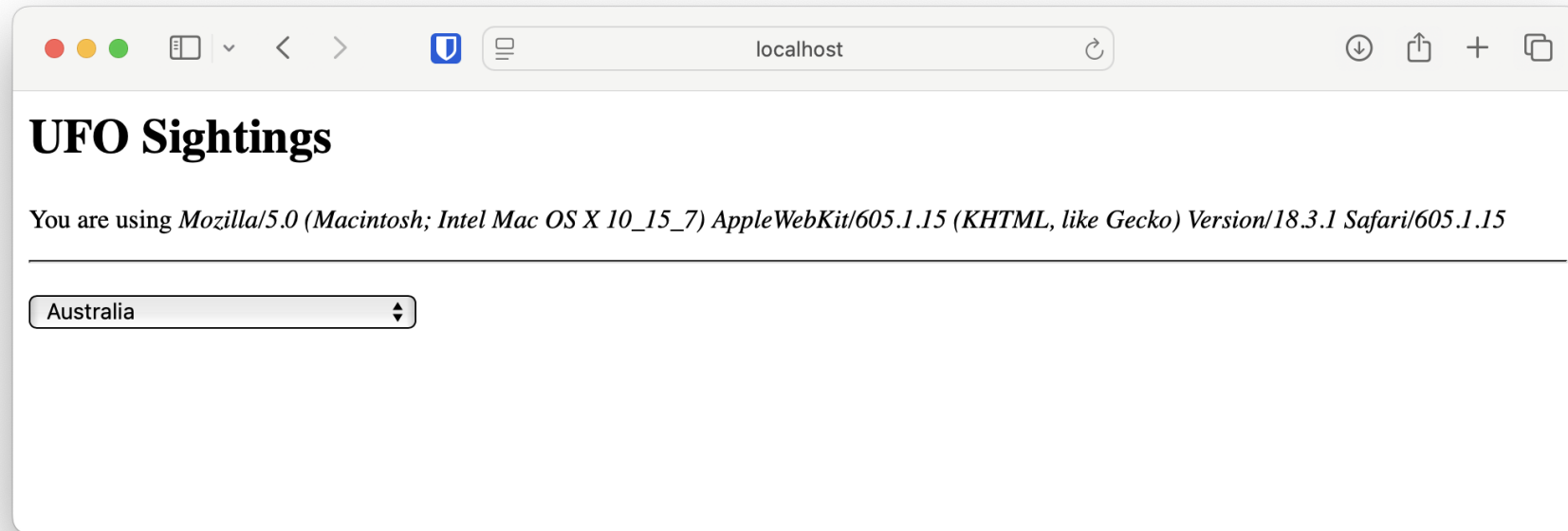


Update index.php file

(...)

```
<hr/>
<select name="Countries" id="Countries">
<?php
    $countries = array(
        "AU" => "Australia",
        "BR" => "Brazil",
        "CA" => "Canada",
        "GB" => "United Kingdom of Great Britain",
        "PT" => "Portugal"
    );
    foreach ($countries as $key => $value) {
        echo "<option value='$key'> $value </option>";
    }
?>
</select>
</body>
</html>
```

Result



- ✓ Australia
- Brazil
- Canada
- United Kingdom of Great Britain
- Portugal

Good Practices

- Create **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php

function getHttpAgent(){
    $browser = $_SERVER['HTTP_USER_AGENT'];
    echo "<p>You are using <i>$browser</i></p>";
}

function setDropBoxOptions() {
    $countries = array(
        "AU" => "Australia",
        (...)
        "PT" => "Portugal"
    );
    foreach ($countries as $key => $value) {
        echo "<option value='$key'> $value </option>";
    }
}

?>
```


Good Practices

- Update the **UFO_Sightings/index.php** file

```
▶ <?php include "BusinessLogicLayer.php"; ?>

<html>
  <head><title>UFO Sightings</title></head>
  <body>
    <h2>UFO Sightings</h2>
    ▶ <?php getHttpAgent(); ?>
    <hr/>
    ▶ <select name="Countries" id="Countries">
      <?php setDropBoxOptions(); ?>
    </select>
  </body>
</html>
```

Get Today's Date

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
```

```
(...)
```

```
function getTodaysDate() {  
    echo "Today is " . Date("d/m/Y") . "<br/>";  
}
```

```
?>
```

- Update the **UFO_Sightings/index.php** file

Send information

- Update the **UFO_Sightings/index.php** file

```
<?php include "BusinessLogicLayer.php"; ?>
<html>
  <head><title>UFO Sightings</title></head>
  <body>
    <div style="text-align:right;width:100%;"><?php getTodaysDate(); ?></div>
    <h2>UFO Sightings</h2>
    <?php getHttpAgent(); ?>
    <hr/>
    <?php if (!isThisACallback()) { ?>
      (...)
    <?php } else { printInfo();} ?>
  </body>
</html>
```

Send information

- Update the **UFO_Sightings/index.php** file

```
(...)  
▶ <?php if (!isThisACallback()) { ?>  
▶     <form action="." method="POST">  
        Date: <input type="date" id="Datetime" name="Datetime"/><br/>  
        City <input type="text" id="City" name="City"/><br/>  
        Country:  
        <select name="Countries" id="Countries">  
            <?php setDropDownOptions(); ?>  
        </select><br/>  
        <p/>  
        <input type="submit" value="Save"/>  
    </form>  
▶ <?php } else { printInfo(); } ?>  
(...)
```

Is a call back?

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
```

```
(...)
```

```
function isThisACallback(): bool{  
    if( empty($_POST['Datetime']) || empty($_POST['City']) || empty($_POST['Countries'])) {  
        return false;  
    }  
    return true;  
}
```

```
(...)
```

```
?>
```

Is a call back! – Print information

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

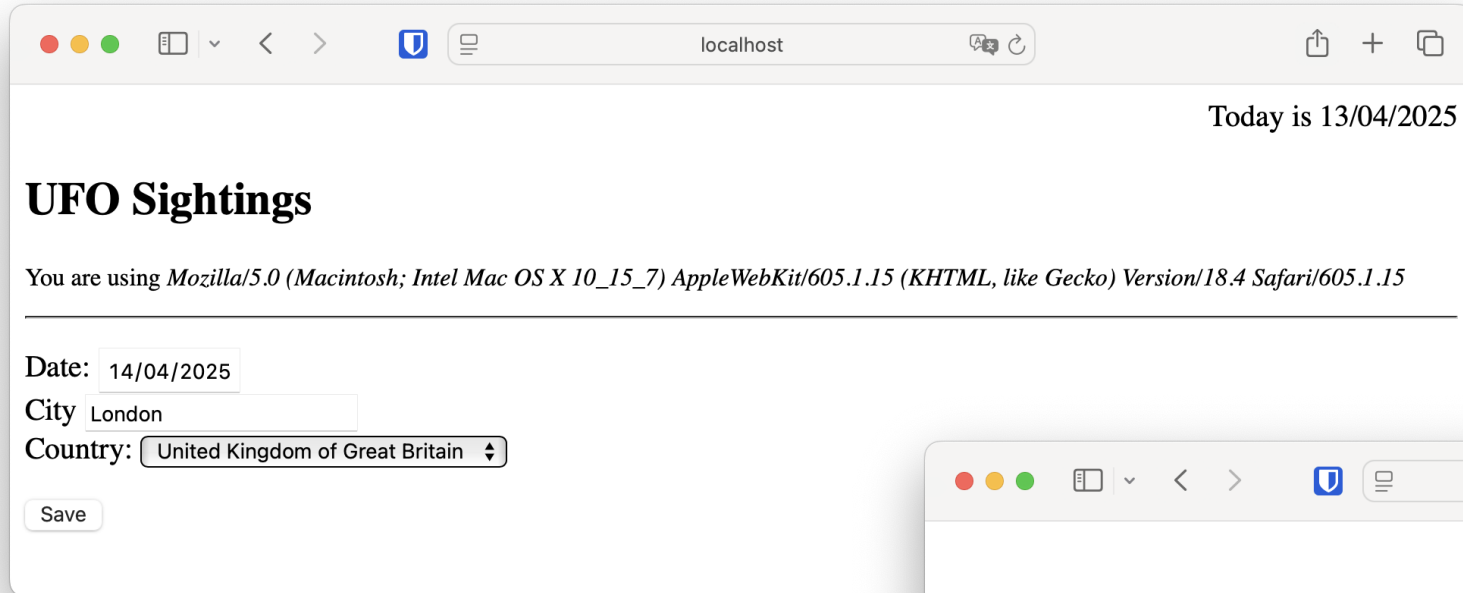
```
<?php
```

```
(...)
```

```
function printInfo() {  
    echo 'Date: ' . $_POST['Datetime'] . '<br/>';  
    echo 'City: ' . htmlspecialchars($_POST['City']) . '<br/>';  
    echo 'Country: ' . $_POST['Countries'] . '<br/>';  
}
```

```
?>
```

Result



A screenshot of a web browser window with the address bar showing 'localhost'. The page title is 'UFO Sightings'. The date 'Today is 13/04/2025' is displayed in the top right. Below the title, a user agent string is shown: 'You are using Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.4 Safari/605.1.15'. The form contains three input fields: 'Date' with the value '14/04/2025', 'City' with the value 'London', and 'Country' with a dropdown menu showing 'United Kingdom of Great Britain'. A 'Save' button is located at the bottom left of the form.

Today is 13/04/2025

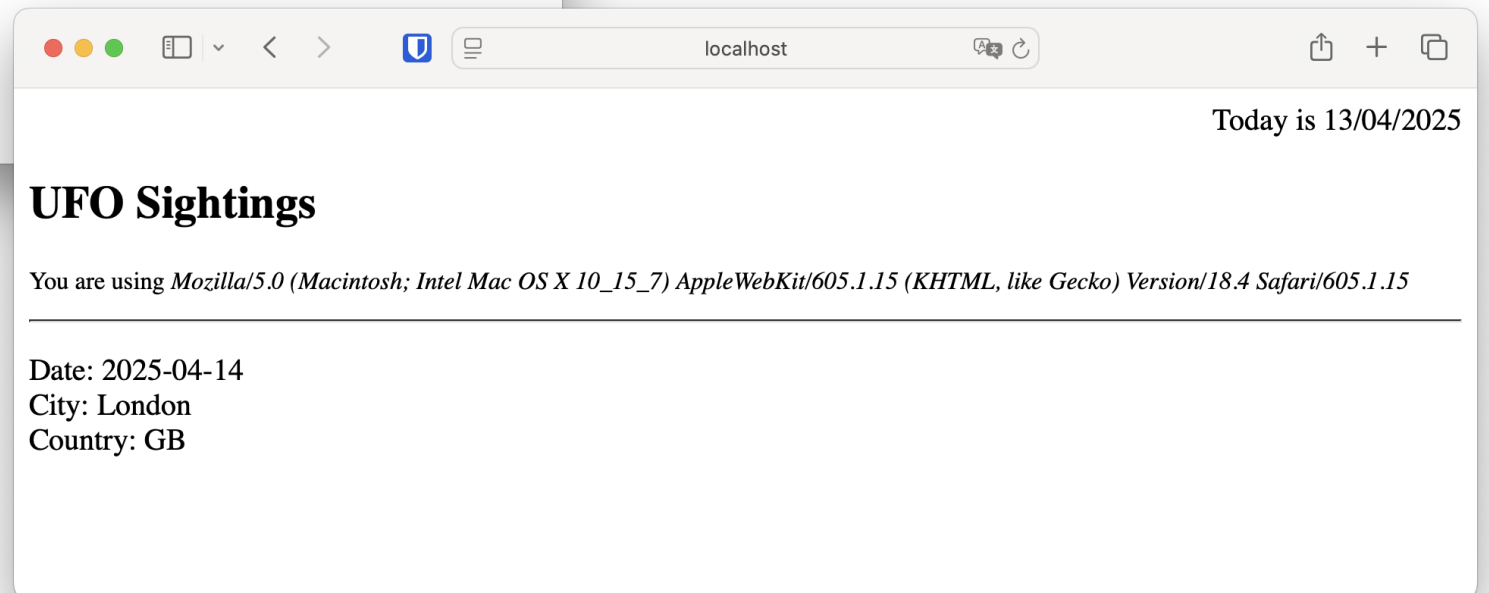
UFO Sightings

You are using *Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.4 Safari/605.1.15*

Date:

City

Country:



A screenshot of a web browser window showing the result of the form submission. The address bar shows 'localhost'. The page title is 'UFO Sightings'. The date 'Today is 13/04/2025' is displayed in the top right. Below the title, the same user agent string is shown. The form fields are now populated with the submitted data: 'Date: 2025-04-14', 'City: London', and 'Country: GB'.

Today is 13/04/2025

UFO Sightings

You are using *Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.4 Safari/605.1.15*

Date: 2025-04-14
City: London
Country: GB

UFO Sightings Information

- Information requirements:

Name	Type	Nullable
datetime	datetime	no
city	varchar(70)	yes
country	varchar(6)	no

Is a call back?

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
```

```
(...)
```

```
function isThisACallback(): bool{  
    if(empty($_POST['Datetime']) || empty($_POST['Countries'])){  
        return false;  
    }  
    return true;  
}
```

```
(...)
```

```
?>
```

Is a call back! – Print information

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
```

```
(...)
```

```
function printInfo() {  
    echo 'Date: ' . $_POST['Datetime'] . '<br/>';  
    if(!empty($_POST['City'])){  
        echo 'City: ' . htmlspecialchars($_POST['City']) . '<br/>';  
    }  
    echo 'Country: ' . $_POST['Countries'] . '<br/>';  
}  
  
?>
```

UFO Sightings Information

- Information requirements:

Name	Type	Nullable
datetime	datetime	no
city	varchar(70)	yes
country	varchar(6)	no
shape	varchar(30)	yes
duration	int	no
comments	text	no
latitude	double	no
longitude	double	no

localhost

Today is 15/04/2025

UFO Sightings

You are using *Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Safari/605.1.15*

Date:

City:

Country:

Shape:

Duration:

Comments:

Latitude:

Longitude:

localhost

Today is 15/04/2025

UFO Sightings

You are using *Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.4 Safari/605.1.15*

Date: 2025-04-15

City: Porto

Country: PT

Shape: Diamond

Duration: 37

Comments: It appeared over the sea near the mouth of the Ouro River. It stabilized about 400 meters above the sea, rocked for 10 seconds and then accelerated vertically upwards.

Latitude: 41.143251

Longitude: -8.677609

Good Practices

- Update the **UFO_Sightings/index.php** file

```
<?php include "BusinessLogicLayer.php"; ?>
<html>
  <head><title>UFO Sightings</title></head>
  <body>
    <div style="text-align:right;width:100%;"><?php getTodaysDate() ?></div>
    <h2>UFO Sightings</h2>
    <?php getHttpAgent(); ?>
    <hr/>
    <p>
      <?php showUI(); ?>
    </p>
  </body>
</html>
```

Good Practices

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
(...)
function displayForm() {
    echo '<form action="." method="POST">
        Date: <input type="date" id="Datetime" name="Datetime"/><br/>
        City: <input type="text" id="City" name="City"/><br/>
        Country: <select name="Countries" id="Countries">';
    setDropBoxOptions();
    echo ' </select><br/>
        Shape: <input type="text" id="Shape" name="Shape"/><br/>
        Duration: <input type="number" id="Duration" name="Duration" min="1" step="1"/><br/>
        Comments: <textarea name="Comments" id="Comments" rows="4" cols="50"></textarea><br/>
        Latitude: <input type="number" id="Latitude" name="Latitude" step="any"/><br/>
        Longitude: <input type="number" id="Longitude" name="Longitude" step="any"/><br/>
        <p></p>
        <input type="submit" value="Save"/>
        </form>';
}
?>
```

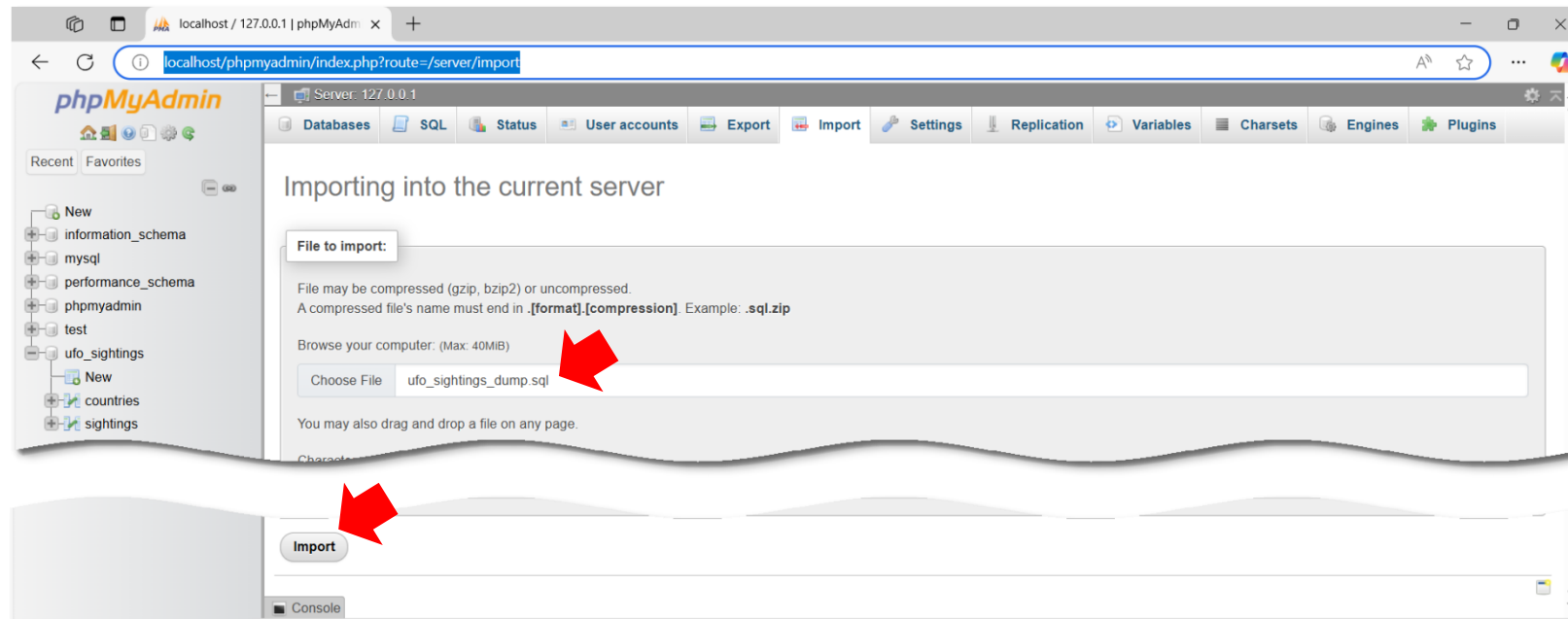
Good Practices

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

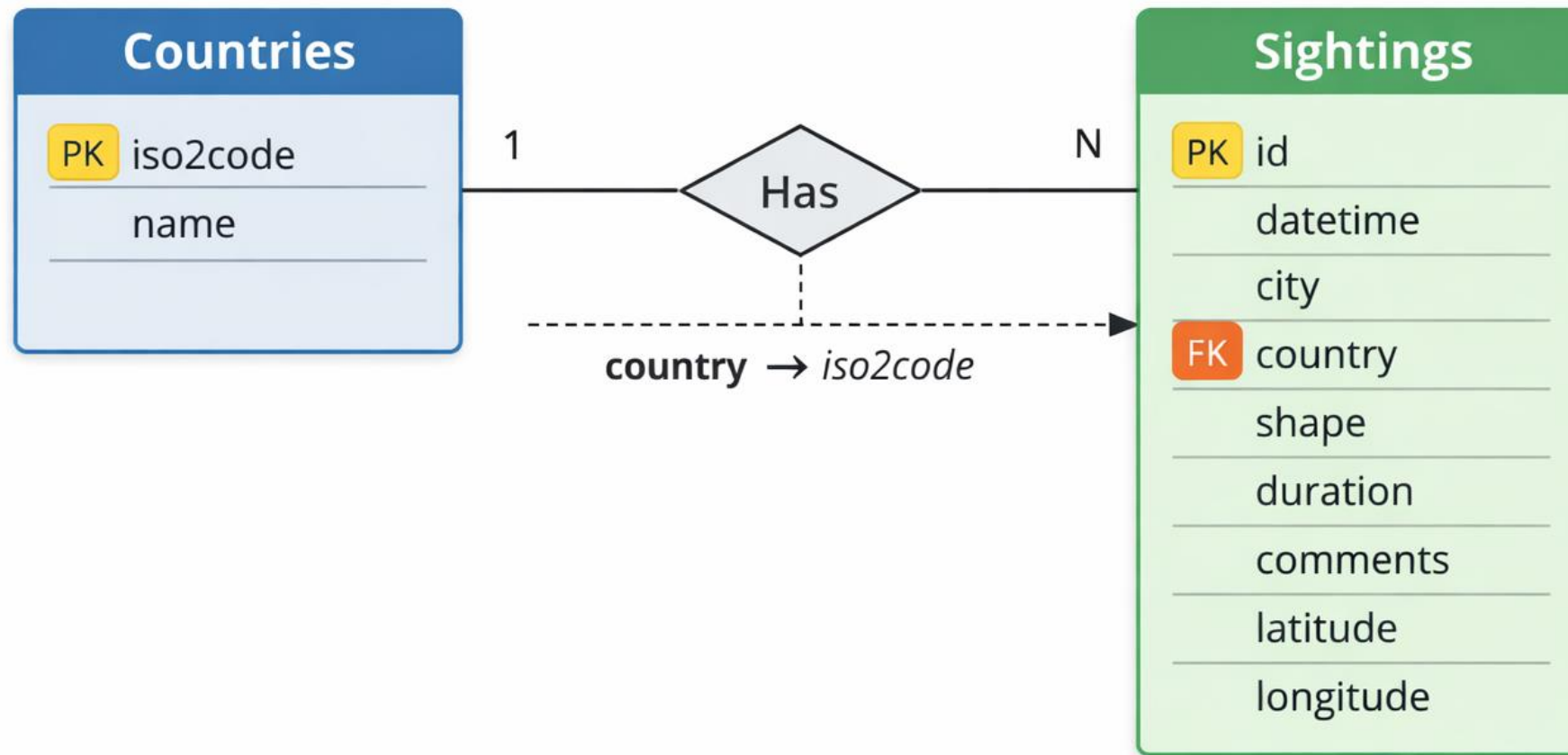
```
<?php  
  
(...)  
  
function showUI(){  
    if(!isThisACallback()){  
        displayForm();  
    }  
    else{  
        printInfo();  
    }  
}  
?>
```

Use the database

- Create the database on the MySQL server:
 - using phpMyAdmin create a database called *ufo_sightings*;
 - download database dump file from moodle;
 - using phpMyAdmin import the dump file to MySQL.



Use the database – ER Model



Use the database

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
▶ include "DataAccessLayer.php";
  (...)
▶ function setDropBoxOptions() {
    try{
        $dal = new DAL();
        $countries = $dal->getAllCountries();
        foreach ($countries as $country) {
            echo "<option value='" . $country["iso2code"] . "' >" . $country["name"] . "</option>";
        }
    }
    catch(RuntimeException $e){
        echo "<option value='ERR'>" . $e->getMessage() . "</option>";
    }
}

(...)
```

Use the database with MySQLi extension

- Create **UFO_Sightings/DataAccessLayer.php** file.

```
<?php
class DAL{
    private $conn;
    function __construct(){
        $this->conn = new mysqli('localhost','root','root','ufo_sightings');
    }
    function getAllCountries(){
        if($this->conn){
            $recordset = $this->conn->query("SELECT * FROM countries");
            $dataset = $recordset->fetch_all(MYSQLI_ASSOC);
            $this->conn->close();
            return $dataset;
        }
        return null;
    }
}
?>
```

Result

Country: Andorra

Shape: Andorra

Duration: United Arab Emirates (the)

Comment: Afghanistan

Latitude: Antigua and Barbuda

Longitude: Anguilla

Save: Albania

Armenia

Angola

Antarctica

Argentina

American Samoa

Austria

Australia

Aruba

Åland Islands

Azerbaijan

Bosnia and Herzegovina

Barbados

Bangladesh

Belgium

Use the database

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
(...)  
function printInfo() {  
    (...)  
    try{  
        $dal = new DAL();  
        $sighting = $dal->createSighting( $_POST['Datetime'],  
            empty($_POST['City']) ? null:htmlspecialchars($_POST['City']),  
            $_POST['Countries'],  
            empty($_POST['Shape']) ? null:htmlspecialchars($_POST['Shape']),  
            $_POST['Duration'], htmlspecialchars($_POST['Comments']),  
            $_POST['Latitude'], $_POST['Longitude'] );  
    }  
    catch(RuntimeException $e){  
        echo "<div>" . $e->getMessage() . "</div>";  
    }  
}  
(...)
```

Use the database with MySQLi extension

- Update **UFO_Sightings/DataAccessLayer.php** file.

```
<?php
class DAL{

    (...)

    function createSighting($date,$city,$country,$shape,$duration,$comment,$latitude,$longitude){
        if($this->conn){
            $theQuery = "INSERT INTO sightings(datetime, city, country, shape, duration, comments,
            latitude, longitude) VALUES (str_to_date('$date','%Y-%m-%d'), '$city', '$country',
            '$shape', $duration, '$comment', $latitude, $longitude)";
            return($this->conn->query($theQuery));
        }
        return false;
    }
}
?>
```

Use the database with PDO extension

- Comment out the **DAL** class in the **DataAccessLayer.php** file.

```
<?php
// class DAL{
//     private $conn;

//     function __construct(){
//         $this->conn = new mysqli('localhost','root','root','ufo_sightings');
//     }

//     function getAllCountries(){
//         (...)
//     }

//     function createSighting($date,$city,$country,$shape,$duration,$comment,$latitude,$longitude){
//         (...)
//     }

// }
?>
```

Use the database with PDO extension

- Update **UFO_Sightings/DataAccessLayer.php** file.

```
<?php
// class DAL{
//     (...)
// }

class DAL{
    private $conn;

    function __construct(){
        $this->conn = new PDO('mysql:host=localhost;dbname=ufo_sightings','root','root');
    }

    (...)
}
?>
```


Use the database with PDO extension

- Update **UFO_Sightings/DataAccessLayer.php** file.

(...)

```
function getAllCountries(){  
    if($this->conn){  
        $statement=$this->conn->query("SELECT * FROM countries");  
        $statement->execute();  
        $dataset = $statement->fetchAll();  
        $this->conn = null;  
        return $dataset;  
    }  
    return null;  
}
```

(...)

Use the database with PDO extension

- Update **UFO_Sightings/DataAccessLayer.php** file.

```
(...)  
function createSighting($date,$city,$country,$shape,$duration,$comment,$latitude,$longitude){  
    if($this->conn){  
        $statement=$this->conn->prepare("INSERT INTO sightings(datetime, city, country, shape,  
duration, comments, latitude, longitude) VALUES (:datetime, :city, :country, :shape, :duration,  
:comments, :latitude, :longitude)");  
        $statement->bindParam(':datetime',$date);  
        $statement->bindParam(':city',$city);  
        $statement->bindParam(':country',$country);  
        $statement->bindParam(':shape',$shape);  
        $statement->bindParam(':duration',$duration);  
        $statement->bindParam(':comments',$comment);  
        $statement->bindParam(':latitude',$latitude);  
        $statement->bindParam(':longitude',$longitude);  
        return($statement->execute());  
    }  
    return false;  
}
```

List all sightings in a country

UFO Sightings

localhost/UFO_Sightings/ListSightings.php

Today is 24/04/2025

UFO Sightings

You are using Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0

Country:

UFO Sightings

localhost/UFO_Sightings/ListSightings.php

Today is 24/04/2025

UFO Sightings

You are using Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0

Date	City	Shape	Duration	Comment
1967-12-14 22:19:21	chester (uk/england)	circle	20	Green/Orange circular disc over Chester, England
1982-08-06 10:50:31	penarth (uk/wales)	circle	180	penarth uk circle 3mins stayed 30ft above me for 3 mins slowly moved of and then with the blink of the eye the speed was unreal
1914-04-28 16:03:10	cardiff (uk/wales)	disk	1200	back in 1974 I was 19 at the time and lived in a suburb of Cardiff Wales UK called Ely, and in the distance there was a wood called Ca
1909-08-				